

Repair What Matters: Criticality-Bounded Revision Cones for Persistent Captured Worlds

SWAYAM SINGAL

Persistent captured-world systems maintain more than a geometric model: they accumulate observations, reconstruction state, explicit surface and appearance representations, GPU-published records, and temporal rendering history. When the physical world changes locally, the corresponding computational revision need not be local, and a spatially restricted update can leave stale dependent state outside the edited region. Rebuilding all state avoids this risk but discards locality when most of the world cannot affect the output of interest.

We present *Criticality-Bounded Revision Cones* (CBRC), implemented in MAVEB, a fail-closed revision framework that selects regional recomputation only when the unrepaired exterior can be conservatively bounded for an explicit quantity of interest. CBRC combines exact predecessor closure for structural dependencies, conservative finite-change bounds for certifiable soft dependencies, a quantity-of-interest error contract, a frozen heterogeneous work model, and an automatic FULL fallback. Every selected result is replayed against an independent full-after reference; the experimental contract is measured full-reference error $\leq$ emitted bound $\leq$ declared tolerance.

In a frozen public campaign of 60 revisions across four RGB/SfM scenes, CBRC selected 44 certified-local repairs and 16 automatic full rebuilds, with zero observed certificate violations and native/reference-planner parity on all 60 cases. Fifty-six of the 60 source edits exceeded the protected RGB tolerance before repair. The frozen calibrated work model reports a median selected work ratio of 0.36953 of FULL (2.706 $\times$ lower calibrated work). A separate five-case validation on a pinned trained 3D Gaussian Splatting representation records four certified-local repairs, one fallback, and zero observed violations. We also identify a remaining systems bottleneck: the frozen end-to-end paths still inspect nearly the full Gaussian set, while a separate exact sparse-discovery study reduces median inspection to 1% without changing candidate selection. These results support certified selective recomputation within the stated assumptions; they do not establish universal safety, global optimality, or end-to-end wall-clock speedup.

CCS Concepts: • **Computing methodologies** $\rightarrow$ **Computer graphics; Rendering; Reconstruction**.

Additional Key Words and Phrases: persistent captured worlds, incremental reconstruction, 3D Gaussian Splatting, certified recomputation, change-aware rendering, dependency analysis

1 INTRODUCTION

Persistent 3D systems are increasingly expected to outlive the capture session that created them. A reconstructed room is revisited after furniture moves; an augmented-reality map persists while objects appear and disappear; a digital twin is refreshed after a local physical intervention. In such systems, the computational state is heterogeneous. Observations feed volumetric or point-based geometry; geometry determines surface or Gaussian support; materials and textures

Author’s address: Swayam Singal, swayam.singal@gmail.com.

depend on ownership and visibility; GPU records mirror persistent entities; temporal filters reuse prior screen-space state. A physical edit that is local in Euclidean space can therefore have a non-local computational footprint.

The classical response is to rebuild globally. Volumetric fusion and dense reconstruction systems already provide a clear full-reference notion of recomputation [5, 6, 15]. Full rebuilding is easy to reason about, but it wastes work when a revision affects only a small dependency closure. Modern explicit scene representations make the waste more visible: Gaussian maps and neural or hybrid reconstructions contain large amounts of state that can often remain unchanged [10, 11, 13]. Continual Gaussian systems consequently restrict optimization to changed regions or manage stale observations over time [2, 23, 24]. These methods establish that local updating is useful, but they answer a different question from the one studied here. They decide *where to update*; they do not generally certify the error permitted to remain outside the chosen region across a heterogeneous captured-world stack.

This paper asks a narrower systems question: *when may a persistent captured world repair less than the whole world while retaining an explicit output-error contract?* The difficulty is not merely reachability. Some dependencies are exact and should never be approximated, such as ownership, record identity, publication, and unstable temporal invalidation. Others have implementation-backed finite influence, such as the display-space contribution of edited Gaussian support or stable temporal blending. A useful revision system must separate these cases, reason about the residual influence outside a proposed repair region, and choose a full rebuild when the residual cannot be bounded tightly enough.

We introduce *Criticality-Bounded Revision Cones* (CBRC). CBRC represents a revision over a typed dependency graph with three edge semantics: HARD edges encode exact structural requirements, ANALYTIC edges carry conservative finite-change upper bounds, and EMPIRICAL edges may guide scheduling but are never used as a safety certificate. For a proposed repair cone C , the remaining exterior O is certified against one or more quantities of interest. We call the worst normalized certificate-to-tolerance ratio the *revision criticality*; the boundary value is one. A local result is accepted only if (1) exact predecessors are closed, (2) revision criticality is at or below one for every protected output, and (3) calibrated regional work is lower than the independently measured full baseline. Unsupported coupling, supercritical residual influence, unstable analytic structure, or an unfavorable work crossover returns FULL.

MAVEB implements the method as a C++23 persistent-world stack. It is paired with a dense Python reference certifier and an independent full-after Gaussian oracle. Figure 1

shows the decision path, and Figure 2 illustrates Gaussian support selection and the two possible outcomes. The implementation records immutable before/after revision sidecars, deterministic certificate JSON, per-domain work, and oracle replay. A certificate is not accepted because the planner predicts that the repair is safe; the selected result is independently replayed against FULL and evaluated under the contract

$$E_q^{\text{full-ref}} \leq B_q(C) \leq \varepsilon_q, \quad (1)$$

where $E_q^{\text{full-ref}}$ is the measured discrepancy from an independent full-after reference, $B_q(C)$ is the emitted conservative bound, and ε_q is the declared tolerance for quantity of interest q .

Our evaluation freezes both protocol and claim boundaries before interpreting the final campaign. Across 60 public RGB/SfM revisions, CBRC selects 44 certified-local repairs and 16 automatic full rebuilds. No certificate violation is observed, native and Python planner decisions agree on all cases, and 56 of 60 source edits exceed the protected RGB tolerance before repair, which rules out the trivial explanation that the campaign contains only visually irrelevant edits. A frozen isolated millisecond cost model reports a median selected heterogeneous work ratio of 0.36953 of FULL, equivalent to $2.706\times$ lower calibrated work. This is a work result, not a paired wall-clock speedup. A second validation on a pinned trained 3D Gaussian Splatting representation records four local selections, one fallback, and zero observed violations. It also exposes the main remaining bottleneck: Gaussian discovery remains nearly global in the frozen end-to-end path.

The contributions are:

- a fail-closed revision formulation for heterogeneous persistent captured worlds that combines exact dependency closure with conservative finite-change propagation;
- quantity-of-interest certification for local repair, including implementation-backed Gaussian display-space and stable temporal bounds, with unsupported dependencies retained as exact;
- a native work-aware revision planner with automatic FULL fallback, deterministic evidence emission, a dense reference implementation, and independent full-after falsification; and
- frozen public and trained-representation evidence that reports local/full crossover, calibrated work, sparse-discovery behavior, negative or neutral ablations, and measured limitations without converting work reduction into speedup claims.

2 RELATED WORK

Persistent reconstruction and map maintenance. Volumetric integration established a durable representation for combining range observations [5], while surfels provided an explicit point-oriented alternative [17]. KinectFusion demonstrated real-time dense fusion and tracking [15]; ElasticFusion maintained and corrected dense surfel maps without a conventional pose graph [19]; and BundleFusion performed globally consistent online surface reintegration [6]. These systems make map state persistent and revisable, but their update logic is tied to representation-specific reconstruction or consistency mechanisms. CBRC instead treats a physical revision as a cross-derived-state decision problem: which state must be recomputed, what residual influence may remain outside that state, and when should incremental execution be abandoned.

Neural and Gaussian mapping. NeRF established continuous neural radiance fields for novel-view synthesis [14]. iMAP and NICE-SLAM brought learned implicit representations into online mapping and SLAM [18, 25]. 3D Gaussian Splatting (3DGS) subsequently provided an explicit anisotropic representation with real-time rendering [11]; SplatAM and Gaussian Splatting SLAM adapted Gaussian representations to online RGB-D and monocular mapping [10, 13]. Dynamic 3D Gaussians and 4D-GS model time-varying scene state [12, 21]. These works motivate explicit revision reasoning because the representation exposes persistent primitives and spatial support, but the CBRC formulation is not specific to Gaussian state.

Continual and evolving Gaussian scenes. CL-Splats segments changed and static scene components to focus local Gaussian optimization and preserve prior scene states [2]. GaussianUpdate performs continual radiance-field updates with change-specific strategies and visibility-aware replay [24]. GaME targets long-term evolving RGB-D scenes and manages stale observations while adapting the Gaussian map [23]. EliGSiR addresses continual Gaussian mapping under a bounded compute budget through adaptive view scheduling, supervision fidelity, and geometry growth [7]. These methods occupy the generic claim that changing Gaussian scenes should be updated locally or selectively. Our claim is different: CBRC couples cross-representation exact dependencies and conservative output bounds, accepts locality only under Eq. 1, and automatically returns FULL when the certificate or work comparison fails.

Editing, change detection, and identity. GaussianEditor and SC-GS show that Gaussian representations support precise or controllable local editing [4, 9]. Recent primitive-space change detection argues that Gaussian attributes can directly reveal structural and appearance change [8]. Consistent Instance Field models persistent probabilistic identity in dynamic Gaussian scenes [22]. These works are important collision classes: local editing, primitive change detection, and persistent identity are not claimed as novel here. CBRC

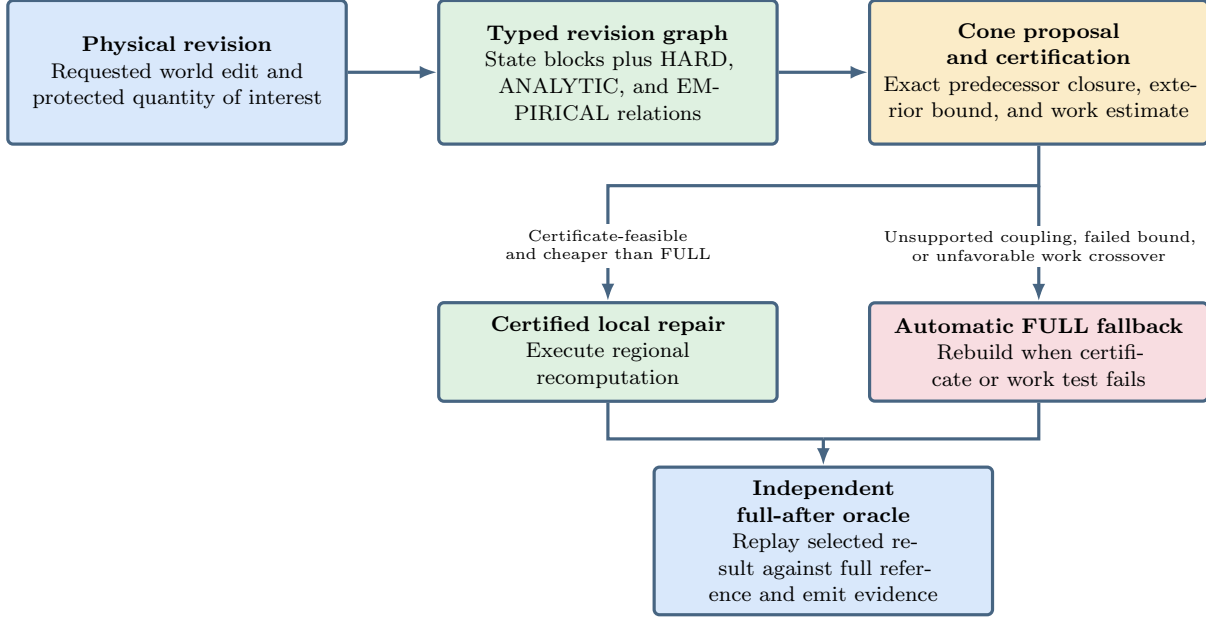


Fig. 1. CBRC decision and falsification path. A physical revision enters a typed dependency graph. Exact HARD predecessors are closed, conservative ANALYTIC influence is propagated to the unrepaired exterior, and quantity-of-interest bounds are checked against tolerance. A local repair is executed only when it is certified and cheaper than the full baseline; otherwise the system falls back to FULL. The selected result is then replayed against an independent full-after oracle.

begins after a revision has been specified or detected and asks whether the consequences outside a candidate repair cone can be bounded for a declared output.

Incremental rendering, change propagation, and goal-oriented error control. Graphics systems have long exploited dependency structure to avoid global recomputation. Bala et al. track dependencies between edited world regions and bounded-error radiance interpolants using ray segment trees [3]. Wörster et al. synthesize a dependency graph from scene-graph changes to rendering caches and update those caches incrementally [20]. Outside graphics, self-adjusting computation formalizes correct change propagation under mutation and reuse [1], while goal-oriented error estimation allocates numerical work to control error in selected quantities of interest rather than a global state norm [16]. These traditions overlap with individual ingredients of CBRC and are therefore important novelty boundaries.

The mathematical tools underlying CBRC remain classical: graph closure, non-negative influence propagation, Neumann-series resolvents, norm bounds, and greedy search are not introduced by this work. The claimed contribution is the captured-world systems contract that combines cross-derived exact dependencies, conservative finite-change bounds, rendered-output quantities of interest, heterogeneous work crossover, fail-closed FULL fallback, and independent full-reference falsification in one revision decision. We therefore treat the mathematics as supporting machinery rather than as standalone novelty.

3 PROBLEM FORMULATION

3.1 Persistent captured-world state

Let the current system state be partitioned into blocks

$$V = \{1, \dots, n\}. \quad (2)$$

A block may represent an observation region, TSDF block, mesh patch, texture page, material state, Gaussian subset, GPU publication range, temporal-history region, or another derived state unit. Blocks are therefore computational ownership units, not necessarily equal-sized spatial cells.

A physical revision induces a direct change envelope $b \in \mathbb{R}_{\geq 0}^n$. The planner chooses a repair cone $C \subseteq V$ and leaves the exterior

$$O = V \setminus C \quad (3)$$

unrepaired. The goal is not to minimize $|C|$. It is to find a certified feasible cone whose effect on the protected output remains inside tolerance and whose measured or calibrated work is lower than the equivalent full rebuild.

3.2 Typed dependencies

Each dependency has one of three semantics (Table 1).

This separation is intentionally conservative. For example, dependencies from observations to TSDF, from TSDF to mesh ownership, from meshes to texture-page identity, and from Gaussian records to GPU publication remain exact. These relations and unstable temporal validation are HARD in the v1 implementation. Gaussian-to-image influence and stable temporal blending have implemented ANALYTIC

How Local Gaussian Revisions Propagate

From intended physical edit to certified local repair or FULL fallback.

MAVEB / CBRC
Certified Local Revision of
Persistent Gaussian Worlds

Edit locally.
Prove the residual.
Fall back to FULL
when necessary.

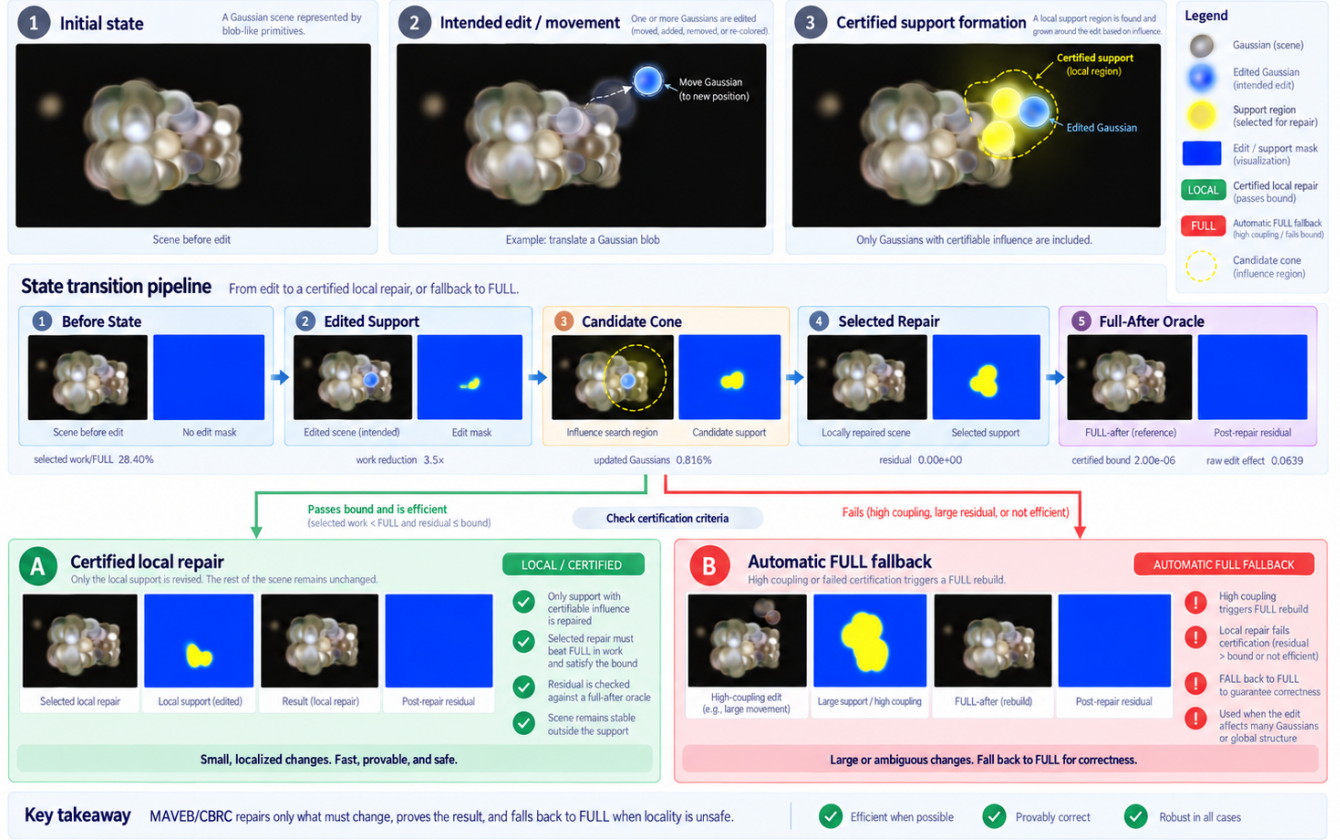


Fig. 2. How a physical edit becomes a certified revision. An intended Gaussian edit induces a candidate influence region from which CBRC selects the support that must be repaired. Exact structural dependencies remain closed and the unrepaired exterior is accepted only when its quantity-of-interest bound satisfies the declared tolerance and regional work remains below the frozen FULL baseline. The system therefore either revises only the certified support (A) or fails closed to a FULL rebuild (B). Blue marks the intended edit, yellow the selected repair support. The supplied image is reproduced unchanged. Its 28.40% work ratio, 3.5 $\times$ work reduction, and other embedded micro-metrics are illustrative, not aggregate campaign results. The frozen public headline is 0.36953 of FULL (2.706 $\times$ lower calibrated work; Section 7). Statements of correctness in the image apply only under the bound assumptions and declared revision domain; neither these work factors nor the image establish wall-clock speedup or universal safety.

Table 1. Revision-edge semantics in CBRC.

Type	Meaning	Certificate role
HARD	Exact ownership, identity, topology, publication, or invalidation dependency	Must be closed exactly
ANALYTIC	Conservative implementation-backed finite-change upper bound	May certify exterior influence
EMPIRICAL	Measured or learned influence used for ordering or scheduling	Never treated as proof

bounds. Relations without a useful conservative theorem are not weakened merely to produce a smaller cone.

3.3 Output and work contracts

For each quantity of interest q , let $\varepsilon_q \geq 0$ be the declared tolerance. A local repair is admissible only if its certified residual satisfies $B_q(C) \leq \varepsilon_q$ for every protected q . The experiment then checks the stronger independent contract in Eq. 1.

The planner also requires regional work to be lower than FULL. Native work counters are heterogeneous, so they are not added directly. If $n_d(C)$ is native work in domain d and

κ_d is a frozen calibrated cost per native unit, the public campaign uses

$$W(C) = \sum_d \kappa_d n_d(C), \quad (4)$$

with an independently measured W_{full} . The calibration is frozen before the final campaign. This distinction matters: Eq. 4 is a work model, not an end-to-end runtime measurement.

4 CRITICALITY-BOUNDED REVISION CONES

4.1 Exact predecessor closure

If a repaired state block v requires an exact predecessor u , then the predecessor must also be inside the cone:

$$v \in C, u \in \text{Pred}_{\text{HARD}}(v) \implies u \in C. \quad (5)$$

This rule handles structural dependencies before any approximate reasoning begins. A candidate violating Eq. 5 is invalid, even if a local numeric sensitivity estimate is small.

4.2 Conservative exterior propagation

For ANALYTIC dependencies, define a componentwise non-negative influence matrix $K \in \mathbb{R}_{\geq 0}^{n \times n}$, where K_{vu} upper-bounds the normalized finite change that block u can transmit to block v . Let z_C denote known change bounds for repaired state. The exterior satisfies

$$\delta_O \leq b_O + K_{OO} z_C + K_{OO} \delta_O. \quad (6)$$

When the exterior response is certifiable and $\rho(K_{OO}) < 1$, the classical resolvent gives

$$G_O = (I - K_{OO})^{-1} = I + K_{OO} + K_{OO}^2 + \dots, \quad (7)$$

and therefore

$$\hat{\delta}_O = G_O (b_O + K_{OC} z_C), \quad \delta_O \leq \hat{\delta}_O. \quad (8)$$

The native v1 planner uses sparse finite DAG propagation rather than general cyclic resolvent evaluation. Unsupported analytic cycles fail closed: the cone must expand until the unsupported exterior is removed, or the planner selects FULL. Equation 8 is therefore the reference formulation, not a claim that arbitrary cyclic systems are handled by the production planner.

4.3 Quantity-of-interest certificate

Let R_q map state perturbation to the protected output quantity. The exterior contribution is bounded by

$$B_q(C) = \left\| |R_{q,O}| \hat{\delta}_O \right\|_{\infty}. \quad (9)$$

A candidate cone is certificate-feasible only if $B_q(C) \leq \varepsilon_q$ for all declared outputs. This allows two revisions with identical changed geometry to produce different decisions when their protected outputs differ. It also prevents an internal state norm from silently standing in for the user-visible error being claimed.

4.4 Revision criticality

We define the revision criticality of a candidate cone as the worst normalized certified residual

$$\chi(C) = \max_q r_q(C),$$

$$r_q(C) = \begin{cases} B_q(C)/\varepsilon_q, & \varepsilon_q > 0, \\ 0, & \varepsilon_q = B_q(C) = 0, \\ +\infty, & \varepsilon_q = 0 < B_q(C). \end{cases} \quad (10)$$

The threshold $\chi(C) = 1$ is the certificate boundary: $\chi(C) \leq 1$ is certificate-feasible, whereas $\chi(C) > 1$ is supercritical for at least one protected output and requires cone expansion or FULL. The term *criticality-bounded* refers to this explicit normalized output-error boundary; it is not intended as a new physical critical-phenomena model.

4.5 Gaussian display-space bound

For a projected Gaussian i at pixel p , MAVEB starts from the projected opacity model

$$\alpha_i(p) = o_i \exp\left(-\frac{1}{2} q_i(p)\right), \quad (11)$$

where o_i is peak opacity and $q_i(p)$ is squared Mahalanobis distance in projected Gaussian space. In the bound below, α_i denotes the effective value after the renderer's clamp and cutoff rules, rather than the raw expression in Eq. 11. For edited set E , define aggregate opacity mass

$$A_p(E) = 1 - \prod_{i \in E} (1 - \alpha_i(p)). \quad (12)$$

Proposition 1 (Gaussian finite-edit display bound). Let U denote the unchanged Gaussian state and E_0, E_1 the complete before/after edited sets at a protected pixel p . Under the assumptions below,

$$\Delta_p = \|R_p(U \cup E_0) - R_p(U \cup E_1)\|_{\infty}, \quad (13)$$

$$\Delta_p \leq C_{\text{color}} \min(1, A_p(E_0) + A_p(E_1)).$$

Assumptions. The bound requires: (i) front-to-back alpha compositing with effective post-clamp/post-cutoff alpha in $[0, 1]$; (ii) each protected color channel lies in an interval of width C_{color} , including the background and the bounded spherical-harmonic evaluation used by the renderer; (iii) E_0, E_1 contain every Gaussian whose projected opacity, color, insertion/deletion state, or depth relation to unchanged support can change because of the edit; and (iv) renderer behavior not represented by these bounded quantities is treated as unsupported and therefore promoted to an exact dependency or FULL fallback.

Proof sketch. For one edited set, front-to-back compositing changes the final channel value by at most the channel range multiplied by the total opacity mass through which edited support can replace, attenuate, or reveal unchanged radiance. This gives $C_{\text{color}} A_p(E)$. Insertion and deletion are the cases in which one edited set is empty. For a general before/after edit, apply the triangle inequality through the common unchanged state U to obtain $C_{\text{color}} (A_p(E_0) + A_p(E_1))$,

then clamp by the maximum possible channel-range change C_{color} . The protected-image certificate is the maximum of Eq. 13 over the declared pixel support. A fuller algebraic derivation is included with the manuscript supplement.

4.6 Stable temporal history

For stable temporal validation, let E_c denote current-frame error, E_h retained-history error, E_n neighborhood or clamping error, and $w \in [0, 1]$ the history weight. With $E_r = \max(E_h, E_n)$,

$$E_{\text{resolved}} \leq (1 - w)E_c + wE_r. \quad (14)$$

If validation, reprojection, or disocclusion may itself change, MAVEB promotes the relation to HARD invalidation instead of applying Eq. 14. Thus the temporal path demonstrates the central design rule: use a soft bound only while its preconditions are certifiable.

4.7 Greedy feasible-cone search and crossover

The v1 search orders candidates using the numerical score $\tilde{\chi}(C) = \max_q B_q(C) / \max(\varepsilon_q, 10^{-15})$ for stable exteriors. This floored score guides search only; acceptance still requires the original inequalities $B_q(C) \leq \varepsilon_q$, including when tolerance is zero. For a candidate expansion $C \rightarrow C'$ with positive added work, it scores

$$\text{utility}(C \rightarrow C') = \frac{\tilde{\chi}(C) - \tilde{\chi}(C')}{W(C') - W(C)}. \quad (15)$$

Every candidate is first closed over exact predecessors and then re-certified. Search stops when a cone is both certificate-feasible and cheaper than FULL, or when no useful local candidate remains. The algorithm claims a certified feasible cone, not the global minimum-work cone.

5 SYSTEM IMPLEMENTATION

MAVEB separates the scientific reference path from the production-oriented execution path. The dense Python certifier implements the mathematical contract directly and is used for falsification and regression. The native C++23 path implements a sparse revision graph with typed nodes and edges, sparse analytic propagation over supported DAG exteriors, exact predecessor consistency, a greedy feasible-cone search, and deterministic certificate serialization. Native and reference-planner decisions are compared in the campaign.

Revision transactions. The production path records persistent Gaussian entity translations, immutable before/after Gaussian revision sidecars, exact camera binding, a production Gaussian image certificate, and planner-selected temporal repair or retention. The headless path loads a persistent world, applies indexed Gaussian transactions, writes a new durable revision and sidecars, evaluates the Gaussian certificate, executes the output-cone planner, and emits canonical evidence JSON. A revision that cannot satisfy consistency checks fails closed rather than publishing partial state.

Cross-derived-state accounting. A **LocalityLedger** is the counter source of truth for observation, TSDF, mesh, texture-page, material, Gaussian, GPU-publication, and temporal domains. Ledger/mesher disagreement is treated as an error. Native counts remain in their own units; conversion to the public scalar work model occurs only through a frozen versioned calibration. This design avoids hiding a full-world stage behind a local metric from a single subsystem.

Independent oracle. The Gaussian oracle is separate from the selected revision planner. It renders or evaluates the equivalent full-after state and reports per-pixel actual error alongside the emitted bound. The strict evaluator enforces Eq. 1; production/offline certificate disagreement or a measured error above the emitted certificate is a correctness failure. Evidence bundles are hashed, and the canonical campaign records workflow IDs, artifact hashes, source hashes, and planner parity.

Representation scope. The public campaign uses four real scenes reconstructed with COLMAP/SfM and seeds isotropic Gaussians from the reconstructed points. These fields are not described as trained photorealistic 3DGS, and public coordinates are normalized rather than metric-scale. A secondary validation path uses a pinned external trained GraphDECO-style PLY while preserving spherical-harmonic coefficients, opacity, anisotropic scale, and rotation. Persistent ownership in this path is deterministic spatial ownership, not semantic segmentation.

6 EVALUATION

6.1 Questions and protocol

The evaluation asks seven questions: (RQ1) does the emitted certificate survive independent full replay; (RQ2) when does the method select LOCAL versus FULL; (RQ3) how much calibrated work is selected relative to FULL; (RQ4) which native work domains remain local; (RQ5) how do simple exact, radius, changed-fraction, and empirical policies compare under the same campaign; (RQ6) does the safety contract transfer to a trained 3DGS representation; and (RQ7) can exact sparse discovery address the observed full-scan Gaussian bottleneck.

The final public v2.1 campaign freezes 60 revisions across Tanks & Temples Train/Truck and Deep Blending Dr Johnson/Playroom inputs distributed in the pinned GraphDECO bundle. The source archive size and SHA-256 are fixed in the repository manifest. The work calibration is frozen before the final campaign. A prior execution in which a requested translation fell below the production effective-edit threshold is preserved as protocol history rather than retuned or removed; v2.1 enforces the production precondition and records requested and applied edit deltas.

For every selected repair, the campaign records the source edit, selected cone, local/full decision, certificate, independent full-reference residual, native work, calibrated work, and

provenance. Fallback cases are retained as results rather than excluded from summary statistics.

Runtime scope. The current campaign does not contain paired end-to-end LOCAL-versus-FULL wall-clock trials under identical execution conditions. Accordingly, no speedup claim is made anywhere in the paper. The public headline is a frozen heterogeneous millisecond-calibrated work ratio, while the baseline suite below reports its own native work-ratio metric. These quantities are intentionally labeled separately.

6.2 Baselines and ablations

The implemented method-level suite contains FULL, exact-only repair, fixed radius policies, a changed-fraction policy, a held-out empirical scheduler, and CBRC. The empirical signal is permitted to order work but is not certificate-eligible. Structural and analytic ablations remove or replace individual components, while a separately labeled deterministic mechanism-isolation suite constructs stress cases for exact closure, analytic Gaussian influence, temporal influence, quantity-of-interest specialization, hard/soft separation, graph-structured propagation, and fallback behavior.

The real campaign is not returned to force every ablation to separate. Several structural ablations are neutral on the frozen matrix; those results are reported as neutral. The synthetic mechanism suite demonstrates that a mechanism can be falsified under a constructed stress case, not that the same mechanism determines effect size on every real revision.

6.3 Metrics

The primary correctness metric is violation count under Eq. 1. We also report local/fallback selection counts, native/reference planner parity, source-edit effect before repair, the median selected calibrated work ratio, and native-domain ratios. Effectivity $\eta = B_q/E_q^{\text{full-ref}}$ is reported only when the independent residual is non-zero; it is undefined for numerically zero residual rather than divided by an arbitrary floor.

7 RESULTS

7.1 Public RGB/SfM campaign

Across the 60 frozen public revisions, CBRC selected a certified local repair in 44 cases and automatically selected FULL in 16. Figure 3 shows representative local and fallback outcomes, while Table 3 summarizes the aggregate evidence. The campaign observed zero certificate violations, and native/reference planner decisions matched on all 60 cases. The maximum measured selected RGB residual was 0 at the reported L_∞ precision; the maximum selected emitted bound was 2×10^{-6} .

Zero selected residual does not imply that the revision matrix was visually trivial. Before repair, 56 of the 60 source edits exceeded the protected RGB tolerance. This source-effect check is important because otherwise a zero post-repair residual could be explained by edits that were already below the quantity-of-interest threshold.

The frozen isolated millisecond cost model reports a median selected heterogeneous work ratio

$$\text{median } W(C)/W_{\text{full}} = 0.3695301, \quad (16)$$

corresponding to $2.706 \times$ lower calibrated work. We intentionally do not call this a $2.706 \times$ speedup: the public scalar is constructed from frozen per-domain microbenchmarks, while campaign phase wall times are recorded separately. A paired local-versus-full end-to-end timing experiment would be required for a speedup claim.

7.2 Method comparison

Table 2 reports the frozen method-level baseline suite on the same 60 cases. The suite re-certifies every selected cone with the conservative certificate, so *certificate pass* is not an independent measured-error metric. In particular, exact-only, radius, and changed-fraction policies often choose very little work but fail the certificate on most cases. The empirical-order baseline uses empirical influence only to order expansion and is still re-certified analytically; the separately held-out changed-fraction heuristic is discussed below.

Table 3. Canonical evidence summary. Work factors are not wall-clock speedups.

	Public RGB/SfM	Trained 3DGS
Cases	60	5
Certified local	44	4
FULL fallback	16	1
Observed violations	0	0
Selected work / FULL	0.36953	0.03495 [†]
Reported reduction	$2.706 \times$	$28.614 \times$ [†]
Median inspection / FULL	≈ 1.0	0.999997

[†]Trained results use native temporal/output work, not the public calibrated model. Ratios and reduction factors are medians.

7.3 Trained-3DGS representation check

The secondary validation uses a pinned public trained 3DGS PLY with spherical-harmonic coefficients, opacity, anisotropic scale, and rotation preserved. Across five frozen revisions, CBRC selected four certified-local repairs and one full fallback with zero observed violations. The median selected native temporal/output work ratio was 0.03495 of FULL, or approximately $28.614 \times$ lower native work under that work definition.

The domain decomposition in Table 4 is more informative than the scalar. Median Gaussian update and GPU publication ratios are approximately 0.0182, and temporal invalidation is approximately 0.03495, but Gaussian inspection remains 0.999997 of FULL. Thus the selected update can be highly local while discovery remains essentially global. We treat this as the principal systems limitation of the frozen end-to-end implementation.

Table 2. Frozen public method comparison. The work column is the baseline suite’s native work-ratio metric and is distinct from the calibrated heterogeneous headline ratio of 0.36953. *Uncertified* means the selected cone failed the conservative certificate, not that an independent oracle violation was observed. All four tested radius policies produced the same aggregate values on this matrix.

Method	Cert. pass	Uncertified	FULL rate	Median native work/FULL	Decision rule
FULL	100.0%	0.0%	100.0%	1.00000	full reference
Exact only	26.7%	73.3%	0.0%	0.00000	exact closure only
Radius 0–3	26.7%	73.3%	0.0%	0.00000	fixed graph radius
Changed fraction	40.0%	60.0%	0.0%	0.00000	threshold policy
Empirical order + certifier	100.0%	0.0%	0.0%	0.04736	empirical ordering, analytic recertification
CBRC	100.0%	0.0%	26.7%	0.04736	certificate + work crossover + fallback

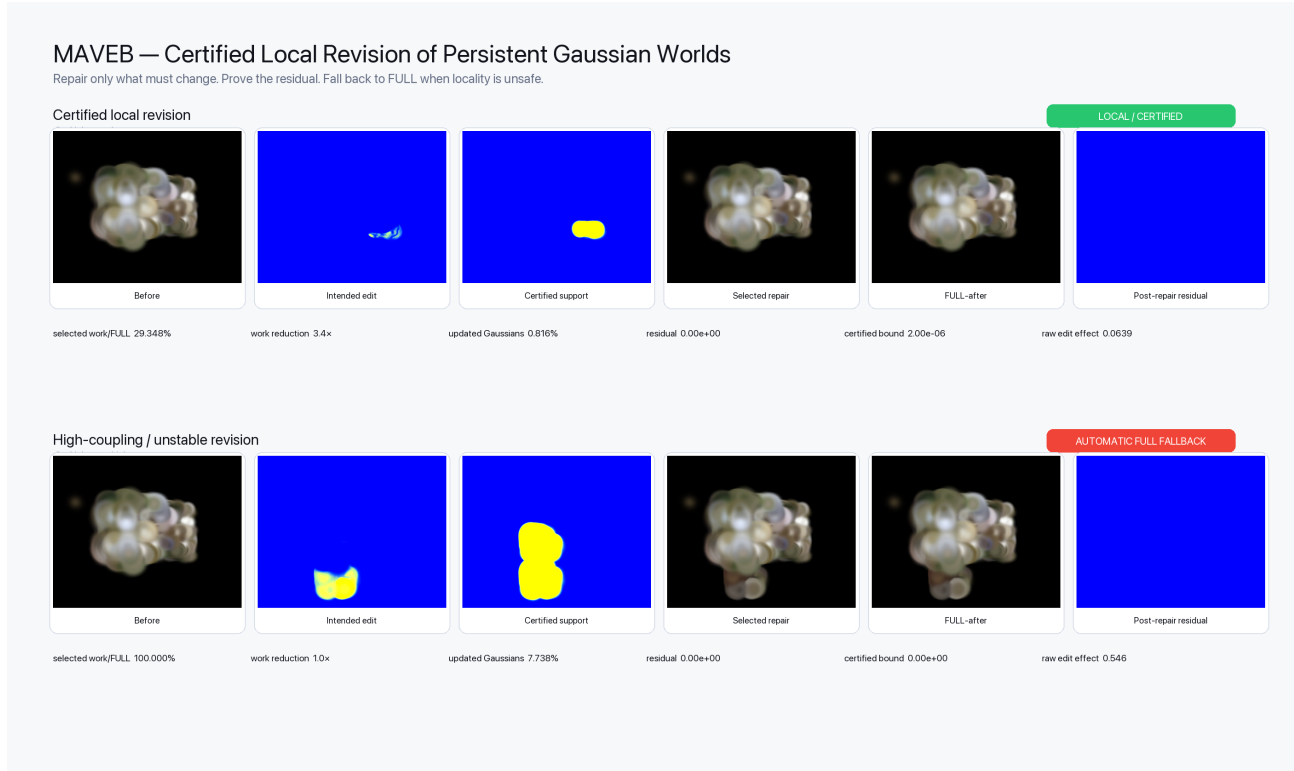


Fig. 3. Representative frozen revision outcomes. Top: a certified local case repairs only the selected support and matches the independent FULL-after reference at the reported precision. Bottom: a higher-coupling revision expands to an automatic FULL fallback rather than forcing a local decision. The intended-edit and certified-support panels make the source effect and selected scope explicit; the residual panel is computed against the independent full-after oracle.

Table 4. Median native-domain ratios in the trained-3DGS validation.

Native domain	Ratio / Full
Gaussian inspection	0.999997
Gaussian update	0.018154
GPU publication bytes	0.018154
Temporal pixels invalidated	0.034948

Inspection remains nearly global even when the selected update, publication, and temporal domains are local. The table exposes rather than hides the discovery bottleneck.

7.4 Sparse candidate discovery

A separate exact sparse-discovery sweep tests whether the near-full inspection requirement is intrinsic to candidate selection. Across the sweep, sparse discovery preserves exact candidate-selection agreement while reducing the median inspected fraction to 0.01 and the minimum to 0.001 through the tested scale of up to one million Gaussians (Table 5). This result demonstrates that the discovery bottleneck is removable in isolation, but it is not silently credited to the frozen end-to-end work ledger because that index was evaluated separately.

Table 5. Separate sparse-discovery study summary.

Statistic	Value
Exact candidate-selection agreement	true
Median inspected fraction	0.010
Minimum inspected fraction	0.001
Largest tested scale	1,000,000 Gaussians

The exact sparse-discovery sweep preserves candidate selection while substantially reducing inspected support. These savings are reported separately and are not credited to the frozen end-to-end headline work number.

7.5 Empirical scheduling and ablations

The held-out changed-fraction scheduler produced no observed unsafe false-local decision on the frozen matrix. This is not evidence that empirical sensitivity is a general safety mechanism: the candidate full-reference residual is zero across these cases, so the matrix does not strongly separate empirical scheduling from certification. CBRC therefore keeps empirical edges outside the certificate by construction.

Table 6 makes the real-matrix ablation behavior explicit. Five structural/analytic ablations are neutral on these 60 cases. The global-norm-tail replacement remains certificate-safe but collapses to FULL on every case. Removing fallback has the opposite failure mode: it never selects FULL and only 26.7% of cases pass the certificate. The separately labeled mechanism-isolation suite contains seven deterministic synthetic stress cases, one for each v1 mechanism, and all seven separate in the expected safety/work direction. Those stress cases establish falsifiability of the mechanisms, not real-world effect size.

The selected public residual is numerically zero in every frozen case at the reported precision, despite 56/60 source edits exceeding tolerance before repair. The real matrix therefore demonstrates selection/fallback correctness but does not estimate certificate tightness under non-zero accepted residual. We use the threshold-crossing mechanism-isolation cases only to show that the analytic machinery can change decisions when stressed; we do not present them as a substitute for real non-zero-residual evidence.

8 DISCUSSION

8.1 What the evidence establishes

Within the frozen revision domain and stated bound assumptions, the experiment supports three concrete statements. First, the emitted certificate survived independent full-reference replay in every evaluated case. Second, the planner did not force locality: more than one quarter of public revisions crossed to FULL. Third, when a local result was selected, the frozen calibrated model often assigned substantially less work than a full rebuild. These statements are stronger than showing that a changed-region heuristic happens to work on average, but weaker than a formal proof of universal production safety.

The most important distinction is between *certification* and *prediction*. An empirical scheduler may correctly guess that a local update is safe on all available cases, yet it remains unable to justify an unobserved case unless its prediction is converted into a conservative bound with a valid remainder. CBRC therefore uses empirical information only for scheduling. Conversely, an ANALYTIC edge may be conservative enough to force FULL frequently; that is a cost of the safety contract, not a correctness failure.

8.2 Limitations

Bounds are only as sound as their assumptions. Every exact dependency needed to reproduce repaired state must be represented by HARD closure, and every ANALYTIC edge must be a finite-change upper bound over the declared revision domain. A representation, shader, temporal rule, or ownership change can invalidate previously calibrated metadata. Certificates therefore require versioned bound provenance and regression tests.

Gaussian discovery remains global in the frozen path. The most direct implementation limitation is the near-full Gaussian inspection ratio. Sparse discovery shows that exact selection can be achieved with far fewer inspections, but integration into the complete measured pipeline remains future engineering. This limitation prevents interpreting the current local update ratios as end-to-end complexity reduction across all phases.

No wall-clock speedup claim. The $2.706\times$ public figure is calibrated heterogeneous work reduction, and the $28.614\times$ trained figure is native temporal/output work reduction. Neither is a paired local-versus-full runtime speedup. A future evaluation should perform matched end-to-end timing under identical execution conditions after sparse discovery is integrated.

Greedy search is not globally optimal. Equation 15 constructs a certified feasible cone but does not prove that no cheaper safe cone exists. Global combinatorial minimum-work optimization is outside the v1 claim.

Analytic cycles fail closed. The reference formulation admits the classical resolvent when its assumptions hold, but the native v1 path certifies finite analytic DAG exteriors. Unsupported cycles trigger cone expansion or FULL. General cyclic certification is future work.

Representation and dataset scope. The primary public campaign uses normalized RGB/SfM reconstructions seeded into Gaussian fields rather than trained photorealistic 3DGS, and it is not a metric-scale reconstruction benchmark. The trained-3DGS experiment is a secondary representation check with only five revisions. Broader trained-scene evaluation would strengthen generality.

Ablation identifiability. Not every mechanism separates on the frozen real matrix. The paper therefore distinguishes real-workload evidence from synthetic mechanism isolation and

Table 6. Frozen real-matrix ablations and their deterministic mechanism-isolation interpretation. Native work ratios use the baseline-suite metric. Neutral means the real matrix matches CBRC on the reported aggregate fields; it is not claimed as evidence that the removed mechanism is unnecessary.

Ablation	Pass	FULL rate	Median native work/FULL	Reading
Remove predecessor closure	100.0%	26.7%	0.04736	neutral on real matrix; synthetic case exposes omitted required exact work
Gaussian analytic → HARD	100.0%	26.7%	0.04736	neutral on real matrix; synthetic case increases required work or forces FULL
Temporal analytic → HARD	100.0%	26.7%	0.04736	neutral on real matrix; synthetic case increases required work or forces FULL
Remove QoI specialization	100.0%	26.7%	0.04736	neutral on real matrix; synthetic case requires extra irrelevant work
Collapse HARD/ANALYTIC separation	100.0%	26.7%	0.04736	neutral on real matrix; synthetic case becomes more conservative
Global-norm exterior tail	100.0%	100.0%	1.00000	safe but destroys locality on all 60 real cases
Remove certified fallback	26.7%	0.0%	0.00000	44/60 real cases fail certification; synthetic case leaves an unsafe candidate

does not claim that each mechanism contributes measurable benefit in every scene.

8.3 Implications

The larger implication is a change in default behavior for persistent captured worlds. A system need not choose between always rebuilding and trusting a local heuristic. If dependencies can be split into exact and conservatively bounded relations, locality can itself become a runtime decision with a falsifiable output contract. This principle extends beyond Gaussian rendering: caches, temporal histories, spatial acceleration structures, compressed representations, and publication buffers can participate so long as their exact or bounded dependency semantics are explicit.

The framework also makes failure scientifically useful. A certificate violation is not averaged into a quality metric; it is retained as a regression target with the revision manifest, graph version, chosen cone, emitted bound, full-reference output, and first violated invariant. This treatment aligns performance evaluation with the correctness burden created by selective recomputation.

9 REPRODUCIBILITY AND ARTIFACT SCOPE

The canonical evidence freeze records workflow run IDs, artifact IDs, Git SHAs, archive hashes, campaign-row hashes, calibration hashes, and the pinned trained-PLY hash. The public source archive is pinned by byte size and SHA-256; the trained PLY is fetched by explicit repository, revision, path, byte size, and SHA-256 and is not redistributed by MAVEB. The project repository includes the dense reference implementation, native planner and oracle, public and trained campaign drivers, baseline and ablation suites, sparse-discovery sweep, deterministic paper-artifact generators, and regression tests.

Scene figures retain applicable upstream terms. Original software is licensed under Apache-2.0; original research communication is scoped under CC BY 4.0; third-party rights are recorded separately. For double-blind review, the public

repository and author metadata should be replaced by an anonymized frozen supplement consistent with venue policy.

10 CONCLUSION

We presented Criticality-Bounded Revision Cones, where criticality is the worst normalized certificate-to-tolerance ratio for the protected outputs, as a fail-closed method for selective recomputation in persistent captured worlds. CBRC separates exact dependencies from conservative finite-change influence, binds the unrepaired exterior to explicit output tolerances, compares regional work against a full baseline, and independently replays every selected result against a full-after oracle. The frozen public and trained-representation campaigns show that local repair and full fallback can coexist under one auditable contract, while also exposing a near-global discovery bottleneck and the absence of a current wall-clock speedup claim. The central principle is simple: a persistent world should repair locally only when it can justify what it leaves untouched; otherwise it should rebuild.

REFERENCES

- [1] Umut A. Acar, Matthias Blume, and Jacob Donham. 2013. A Consistent Semantics of Self-Adjusting Computation. *Journal of Functional Programming* 23, 3 (2013), 249–292. <https://doi.org/10.1017/S0956796813000099>
- [2] Jan Ackermann, Jonas Kulhanek, Shengqu Cai, Haofei Xu, Marc Pollefeys, Gordon Wetzstein, Leonidas J. Guibas, and Songyou Peng. 2025. CL-Splats: Continual Learning of Gaussian Splatting with Local Optimization. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 7808–7817.
- [3] Kavita Bala, Julie Dorsey, and Seth Teller. 1999. Interactive Ray-Traced Scene Editing Using Ray Segment Trees. In *Eurographics Workshop on Rendering*. 31–44. <https://doi.org/10.2312/EGWR/EGWR99/031-044>
- [4] Yiwen Chen, Zilong Chen, Chi Zhang, Feng Wang, Xiaofeng Yang, Yikai Wang, Zhongang Cai, Lei Yang, Huaping Liu, and Guosheng Lin. 2024. GaussianEditor: Swift and Controllable 3D Editing with Gaussian Splatting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 21476–21485. <https://doi.org/10.1109/CVPR52733.2024.02029>
- [5] Brian Curless and Marc Levoy. 1996. A Volumetric Method for Building Complex Models from Range Images. In *Proceedings of SIGGRAPH*. 303–312.
- [6] Angela Dai, Matthias Nießner, Michael Zollhöfer, Shahram Izadi, and Christian Theobalt. 2017. BundleFusion: Real-Time Globally

- Consistent 3D Reconstruction Using On-the-Fly Surface Reintegration. *ACM Transactions on Graphics* 36, 4 (2017), 76:1–76:18.
- [7] Björn Ellensohn, Elmar Rueckert, and Christian Rauch. 2026. EliGSiR: Continual RGB-D Mapping with Gaussian Splatting under Bounded Compute. *arXiv preprint arXiv:2609.20348* (2026).
 - [8] Chamuditha Jayanga Galappaththige, Jason Lai, Timothy Patten, Donald Dansereau, Niko Suenderhauf, and Dimity Miller. 2026. From Pixels to Primitives: Scene Change Detection in 3D Gaussian Splatting. *arXiv preprint arXiv:2605.07203* (2026).
 - [9] Yi-Hua Huang, Yang-Tian Sun, Ziyi Yang, Xiaoyang Lyu, Yan-Pei Cao, and Xiaojuan Qi. 2024. SC-GS: Sparse-Controlled Gaussian Splatting for Editable Dynamic Scenes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 4220–4230. <https://doi.org/10.1109/CVPR52733.2024.00404>
 - [10] Nikhil Keetha, Jay Karhade, Krishna Murthy Jatavallabhula, Gengshan Yang, Sebastian Scherer, Deva Ramanan, and Jonathon Luiten. 2024. SplatTAM: Splat Track & Map 3D Gaussians for Dense RGB-D SLAM. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 21357–21366.
 - [11] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 2023. 3D Gaussian Splatting for Real-Time Radiance Field Rendering. *ACM Transactions on Graphics* 42, 4 (July 2023). <https://doi.org/10.1145/3592433>
 - [12] Jonathon Luiten, Georgios Kopanas, Bastian Leibe, and Deva Ramanan. 2024. Dynamic 3D Gaussians: Tracking by Persistent Dynamic View Synthesis. In *International Conference on 3D Vision*. <https://doi.org/10.1109/3DV62453.2024.00044>
 - [13] Hidenobu Matsuki, Riku Murai, Paul H. J. Kelly, and Andrew J. Davison. 2024. Gaussian Splatting SLAM. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 18039–18048. <https://doi.org/10.1109/CVPR52733.2024.01708>
 - [14] Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. 2020. NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis. In *European Conference on Computer Vision*. 405–421.
 - [15] Richard A. Newcombe, Shahram Izadi, Otmar Hilliges, David Molyneaux, David Kim, Andrew J. Davison, Pushmeet Kohli, Jamie Shotton, Steve Hodges, and Andrew Fitzgibbon. 2011. KinectFusion: Real-Time Dense Surface Mapping and Tracking. In *IEEE International Symposium on Mixed and Augmented Reality*. 127–136.
 - [16] J. Tinsley Oden and Serge Prudhomme. 2001. Goal-Oriented Error Estimation and Adaptivity for the Finite Element Method. *Computers & Mathematics with Applications* 41, 5–6 (2001), 735–756. [https://doi.org/10.1016/S0898-1221\(00\)00317-5](https://doi.org/10.1016/S0898-1221(00)00317-5)
 - [17] Hanspeter Pfister, Matthias Zwicker, Jeroen van Baar, and Markus Gross. 2000. Surfels: Surface Elements as Rendering Primitives. In *Proceedings of SIGGRAPH*. 335–342. <https://doi.org/10.1145/344779.344936>
 - [18] Edgar Sucar, Shikun Liu, Joseph Ortiz, and Andrew J. Davison. 2021. iMAP: Implicit Mapping and Positioning in Real-Time. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 6229–6238. <https://doi.org/10.1109/ICCV48922.2021.00617>
 - [19] Thomas Whelan, Stefan Leutenegger, Renato F. Salas-Moreno, Ben Glocker, and Andrew J. Davison. 2015. ElasticFusion: Dense SLAM Without a Pose Graph. In *Robotics: Science and Systems*.
 - [20] Michael Wörister, Harald Steinlechner, Stefan Maierhofer, and Robert F. Tobler. 2013. Lazy Incremental Computation for Efficient Scene Graph Rendering. In *Eurographics/ACM SIGGRAPH Symposium on High Performance Graphics*. 53–62. <https://doi.org/10.1145/2492045.2492051>
 - [21] Guanjun Wu, Taoran Yi, Jiemin Fang, Lingxi Xie, Xiaopeng Zhang, Wei Wei, Wenyu Liu, Qi Tian, and Xinggang Wang. 2024. 4D Gaussian Splatting for Real-Time Dynamic Scene Rendering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 20310–20320. <https://doi.org/10.1109/CVPR52733.2024.01920>
 - [22] Junyi Wu, Van Nguyen Nguyen, Benjamin Planche, Jiachen Tao, Changchang Sun, Zhongpai Gao, Zhenghao Zhao, Anwesa Choudhuri, Gengyu Zhang, Meng Zheng, Feiran Wang, Terrence Chen, Yan Yan, and Ziyang Wu. 2026. Consistent Instance Field for Dynamic Scene Understanding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 3783–3793.
 - [23] Vladimir Yugay, Thies Kersten, Luca Carlone, Theo Gevers, Martin R. Oswald, and Lukas Schmid. 2026. Gaussian Mapping for Evolving Scenes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 18903–18912.
 - [24] Lin Zeng, Boming Zhao, Jiarui Hu, Xujie Shen, Ziqiang Dang, Hujun Bao, and Zhaopeng Cui. 2025. GaussianUpdate: Continual 3D Gaussian Splatting Update for Changing Environments. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 25800–25809. <https://doi.org/10.1109/ICCV51701.2025.02393>
 - [25] Zihan Zhu, Songyou Peng, Viktor Larsson, Weiwei Xu, Hujun Bao, Zhaopeng Cui, Martin R. Oswald, and Marc Pollefeys. 2022. NICE-SLAM: Neural Implicit Scalable Encoding for SLAM. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 12786–12796.